ADVANCED

Stage 6

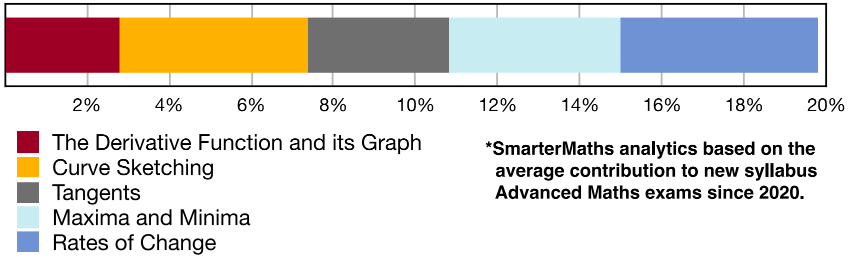
Calculus (Adv), C3 Applications of Calculus (Adv)

Maxima and Minima Part 2 (Y12)

Teacher: SHS

Exam Equivalent Time: 142.5 minutes *(based on allocation of 1.5 minutes per mark)*

C3 Applications of Calculus



***SmarterMaths analytics based on the average contribution to new syllabus Advanced Maths exams since 2020.**

HISTORICAL CONTRIBUTION

- *C3 Applications of Calculus* is the 800 pound gorilla in the Advanced rainforest, contributing a massive 19.8% to new syllabus exams since introduced in 2020.
- This topic has been split into five sub-topics for analysis purposes: *1-The Derivative Function and its Graph (2.8%), 2-Curve Sketching (4.6%), 3-Tangents (3.4%), 4-Maxima and Minima (4.2%) and 5-Rates of Change (4.8%).*
- This analysis looks at *Maxima and Minima.*

HSC ANALYSIS - What to expect and common pitfalls

- *Maxima and Minima (4.2%)* appears in every exam with significant mark allocations almost guaranteed. Its importance in achieving a band 5-6 result cannot be overstated.
- *Max/Min* problems will typically ask students to prove an equation in the earlier parts. A band 4 level task that presents a good opportunity for all students to score well.
- HSC Markers remind students to work with the given equation of the question, *even if they couldn't successfully complete the proof in the earlier part.*
- *Max/Min* problems have had a number of specific themes. *Area/Volume equations* have dominated, appearing in 4 out of 5 new syllabus exams. Other themes include *distance* (2017 and 2013) and a *cost equation* (2009).
- *Max/Min* was novelly examined within the *Probability Density Function* topic in 2021 and is included in the database under that topic.

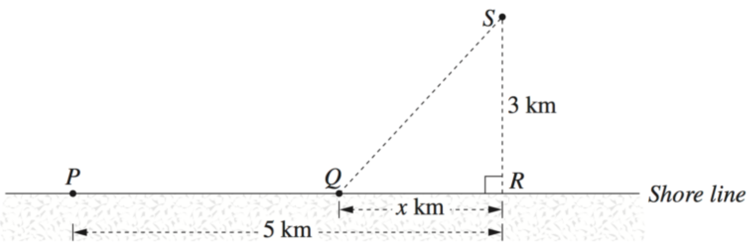
Questions

1. Calculus, 2ADV C3 2009 HSC 9b

An oil rig, S, is 3 km offshore. A power station, P, is on the shore. A cable is to be laid from P to S. It costs \$1000 per kilometre to lay the cable along the shore and \$2600 per kilometre to lay the cable underwater from the shore to S.

The point R is the point on the shore closest to S, and the distance PR is 5 km.

The point Q is on the shore, at a distance of x km from R, as shown in the diagram.



i. Find the total cost of laying the cable in a straight line from P to R and then in a straight line from R to S. *(1 mark)*

ii. Find the cost of laying the cable in a straight line from P to S. *(1 mark)*

iii. Let $\$C$ be the total cost of laying the cable in a straight line from P to Q, and then in a straight line from Q to S.

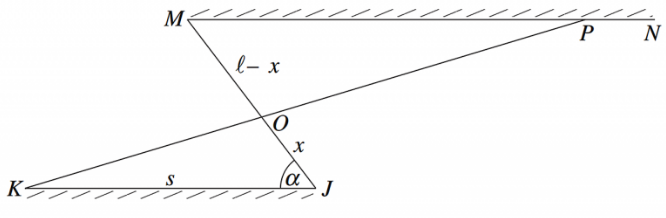
Show that $C = 1000\left(5 - x + 2.6\sqrt{x^2+9}\right)$. *(2 marks)*

iv. Find the minimum cost of laying the cable. *(4 marks)*

v. New technology means that the cost of laying the cable underwater can be reduced to \$1100 per kilometre.

Determine the path for laying the cable in order to minimise the cost in this case. *(2 marks)*

2. Calculus, 2ADV C3 2008 HSC 10b



The diagram shows two parallel brick walls KJ and MN joined by a fence from J to M. The wall KJ is s metres long and $\angle KJM = \alpha$. The fence JM is l metres long.

A new fence is to be built from K to a point P somewhere on MN. The new fence KP will cross the original fence JM at O.

Let $OJ = x$ metres, where $0 < x < l$.

i. Show that the total area, A square metres, enclosed by ΔOKJ and ΔOMP is given by

$$A = s\left(x - l + \frac{l^2}{2x}\right)\sin\alpha.$$ *(3 marks)*

ii. Find the value of x that makes A as small as possible. Justify the fact that this value of x gives the minimum value for A. *(3 marks)*

iii. Hence, find the length of MP when A is as small as possible. *(1 mark)*

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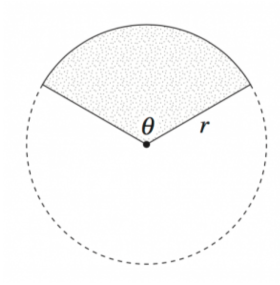
3. Calculus, 2ADV C3 2011 HSC 10b

A farmer is fencing a paddock using P metres of fencing. The paddock is to be in the shape of a sector of a circle with radius r and sector angle θ in radians, as shown in the diagram.



i. Show that the length of fencing required to fence the perimeter of the paddock is

$$P = r(\theta + 2).$$ *(1 mark)*

ii. Show that the area of the sector is $A = \frac{1}{2}Pr - r^2$. *(1 mark)*

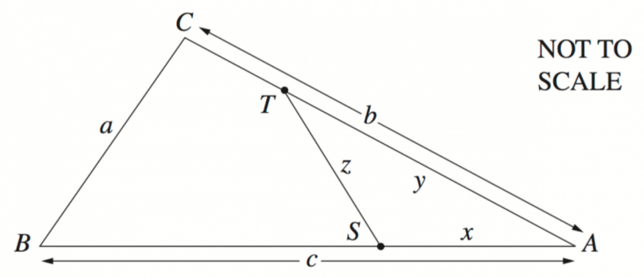
iii. Find the radius of the sector, in terms of P, that will maximise the area of the paddock. *(2 marks)*

iv. Find the angle θ that gives the maximum area of the paddock. *(1 mark)*

v. Explain why it is only possible to construct a paddock in the shape of a sector if

$$\frac{P}{2(\pi + 1)} < r < \frac{P}{2}$$ *(2 marks)*

4. Calculus, 2ADV C3 2004 HSC 10b



The diagram shows a triangular piece of land ABC with dimensions $AB = c$ metres, $AC = b$ metres and $BC = a$ metres, where $a \leq b \leq c$.

The owner of the land wants to build a straight fence to divide the land into two pieces of equal area. Let S and T be points on AB and AC respectively so that ST divides the land into two pieces of equal area.

Let $AS = x$ metres, $AT = y$ metres and $ST = z$ metres.

i. Show that $xy = \frac{1}{2}bc$. *(1 mark)*

ii. Use the cosine rule in triangle AST to show that

$$z^2 = x^2 + \frac{b^2c^2}{4x^2} - bc\cos A.$$ *(2 marks)*

iii. Show that the value of z^2 in the equation in part (ii) is a minimum when

$$x = \sqrt{\frac{bc}{2}}.$$ *(4 marks)*

iv. Show that the minimum length of the fence is $\sqrt{\dfrac{(P-2b)(P-2c)}{2}}$ metres, where $P = a + b + c$.

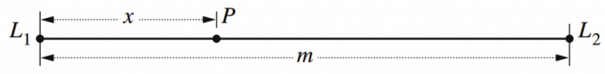
(You may assume that the value of x given in part (iii) is feasible.) *(2 marks)*

5. Calculus, 2ADV C3 2007 HSC 10b

The noise level, N, at a distance d metres from a single sound source of loudness L is given by the formula

$$N = \frac{L}{d^2}.$$

Two sound sources, of loudness L_1 and L_2 are placed m metres apart.



The point P lies on the line between the sound sources and is x metres from the sound source with loudness L_1.

i. Write down a formula for the sum of the noise levels at P in terms of x. *(1 mark)*

ii. There is a point on the line between the sound sources where the sum of the noise levels is a minimum. Find an expression for x in terms of m, L_1 and L_2 if P is chosen to be this point. *(4 marks)*

6. Calculus, 2ADV C3 2017 HSC 16a

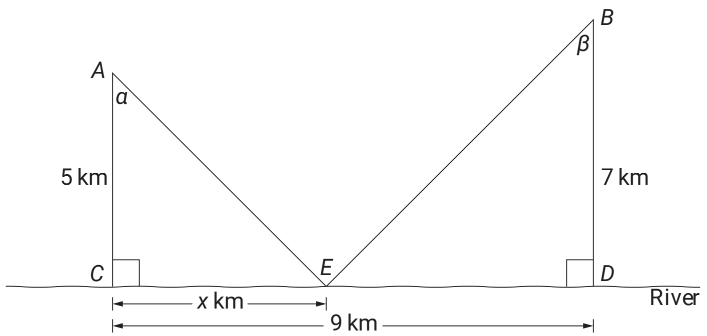
John's home is at point A and his school is at point B. A straight river runs nearby.

The point on the river closest to A is point C, which is 5 km from A.

The point on the river closest to B is point D, which is 7 km from B.

The distance from C to D is 9 km.

To get some exercise, John cycles from home directly to point E on the river, x km from C, before cycling directly to school at B, as shown in the diagram.



The total distance John cycles from home to school is L km.

i. Show that $L = \sqrt{x^2 + 25} + \sqrt{49 + (9 - x)^2}$. *(1 mark)*

ii. Show that if $\dfrac{dL}{dx} = 0$, then $\sin\alpha = \sin\beta$. *(3 marks)*

iii. Find the value of x that makes $\sin\alpha = \sin\beta$. *(2 marks)*

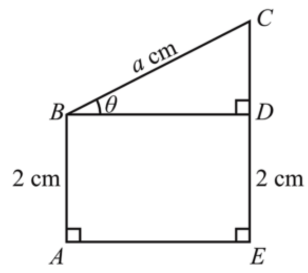
iv. Explain why this value of x gives a minimum for L. *(1 mark)*

7. Calculus, 2ADV C3 SM-Bank 13

The figure shown represents a wire frame where $ABCE$ is a convex quadrilateral. The point D is on line segment EC with $AB = ED = 2$ cm and $BC = a$ cm, where a is a positive constant.

$$\angle BAE = \angle CEA = \frac{\pi}{2}$$

Let $\angle CBD = \theta$ where $0 < \theta < \dfrac{\pi}{2}$.



i. Find BD and CD in terms of a and θ. *(2 marks)*

ii. Find the length, L cm, of the wire in the frame, including length BD, in terms of a and θ. *(1 mark)*

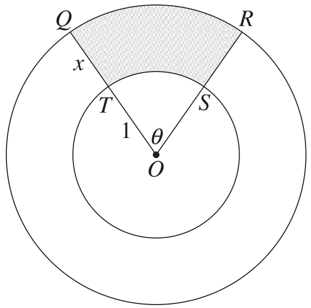
iii. Find $\dfrac{dL}{d\theta}$, and **hence** show that $\dfrac{dL}{d\theta} = 0$ when $BD = 2CD$. *(2 marks)*

iv. Find the maximum value of L if $a = 3\sqrt{5}$. *(1 mark)*

8. Calculus, 2ADV C3 2024 HSC 31

Two circles have the same centre O. The smaller circle has radius 1 cm, while the larger circle has radius $(1 + x)$ cm. The circles enclose a region $QRST$, which is subtended by an angle θ at O, as shaded.

The area of $QRST$ is A cm^2, where A is a constant and $A > 0$.



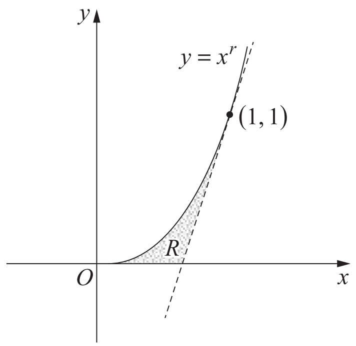
Let P cm be the perimeter of $QRST$.

a. By finding expressions for the area and perimeter of $QRST$, show that $P(x) = 2x + \dfrac{2A}{x}$. *(3 marks)*

b. Show that if the perimeter, $P(x)$, is minimised, then θ must be less than 2. *(3 marks)*

9. Calculus, 2ADV C4 2019 HSC 16c

The diagram shows the region R, bounded by the curve $y = x^r$, where $r \geq 1$, the x-axis and the tangent to the curve at the point $(1, 1)$.



i. Show that the tangent to the curve at $(1, 1)$ meets the x-axis at

$\left(\frac{r-1}{r}, 0\right)$. ***(2 marks)***

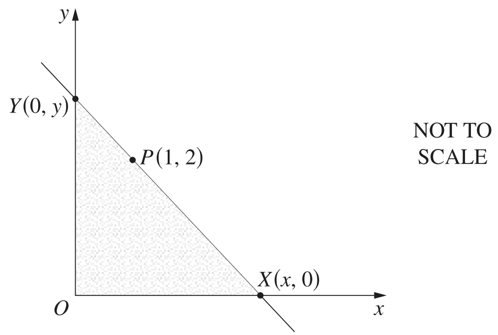
ii. Using the result of part (i), or otherwise, show that the area of the region R is

$\frac{r-1}{2r(r+1)}$. ***(2 marks)***

iii. Find the exact value of r for which the area of R is a maximum. ***(3 marks)***

10. Calculus, 2ADV C3 2022 HSC 31

A line passes through the point $P(1, 2)$ and meets the axes at $X(x, 0)$ and $Y(0, y)$, where $x > 1$.



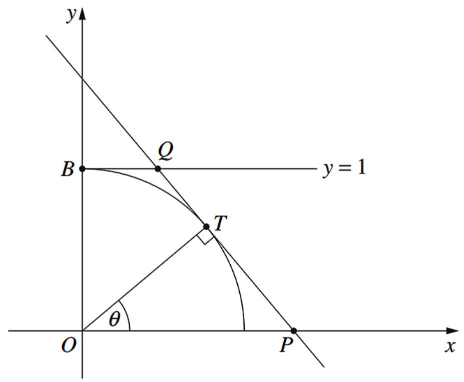
a. Show that $y = \frac{2x}{x-1}$. ***(2 marks)***

b. Find the minimum value of the area of triangle XOY. ***(4 marks)***

11. Calculus, 2ADV C3 2012 HSC 16b

The diagram shows a point T on the unit circle $x^2 + y^2 = 1$ at an angle θ from the positive x-axis, where $0 < \theta < \frac{\pi}{2}$.

The tangent to the circle at T is perpendicular to OT, and intersects the x-axis at P, and the line $y = 1$ intersects the y-axis at B.



i. Show that the equation of the line PT is $x\cos\theta + y\sin\theta = 1$. ***(2 marks)***

ii. Find the length of BQ in terms of θ. ***(1 mark)***

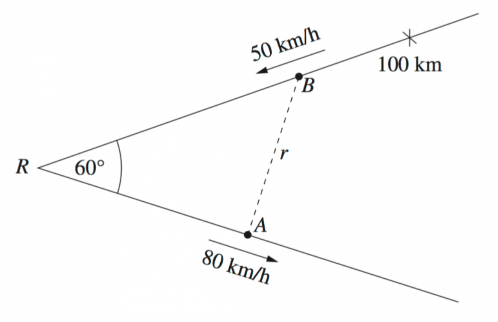
iii. Show that the area, A, of the trapezium $OPQB$ is given by

$$A = \frac{2 - \sin\theta}{2\cos\theta}$$ ***(2 marks)***

iv. Find the angle θ that gives the minimum area of the trapezium. ***(3 marks)***

12. Calculus, 2ADV C3 2013 HSC 14b

Two straight roads meet at R at an angle of 60°. At time $t=0$ car A leaves R on one road, and car B is 100km from R on the other road. Car A travels away from R at a speed of 80 km/h, and car B travels towards R at a speed of 50 km/h.



The distance between the cars at time t hours is r km.

i. Show that $r^2 = 12\ 900t^2 - 18\ 000t + 10\ 000$. ***(2 marks)***

ii. Find the minimum distance between the cars. ***(3 marks)***

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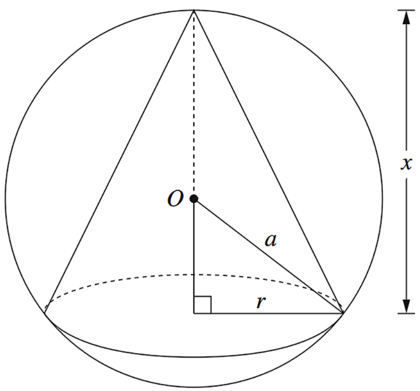
13. Calculus, 2ADV C3 2006 HSC 9c



A cone is inscribed in a sphere of radius a, centred at O. The height of the cone is x and the radius of the base is r, as shown in the diagram.

i. Show that the volume, V, of the cone is given by

$$V = \frac{1}{3}\pi\left(2ax^2 - x^3\right).$$ ***(2 marks)***

ii. Find the value of x for which the volume of the cone is a maximum. You must give reasons why your value of x gives the maximum volume. ***(3 marks)***

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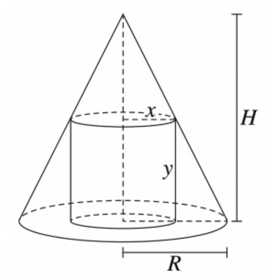
14. Calculus, 2ADV C3 2015 HSC 16c

The diagram shows a cylinder of radius x and height y inscribed in a cone of radius R and height H, where R and H are constants.



The volume of a cone of radius r and height h is $\frac{1}{3}\pi r^2 h$.

The volume of a cylinder of radius r and height h is $\pi r^2 h$.

i. Show that the volume, V, of the cylinder can be written as

$$V = \frac{H}{R}\pi x^2 (R - x).$$ ***(3 marks)***

ii. By considering the inscribed cylinder of maximum volume, show that the volume of any inscribed cylinder does not exceed $\frac{4}{9}$ of the volume of the cone. ***(4 marks)***

Worked Solutions

1. Calculus, 2ADV C3 2009 HSC 9b

♦♦ *Although specific data is unavailable for question parts, mean marks were 35% for Q9 in total.*

i.
$$\begin{aligned}\text{Cost} &= (PR \times 1000) + (SR \times 2600)\\ &= (5 \times 1000) + (3 \times 2600)\\ &= 12\ 800\end{aligned}$$

$\therefore$ Cost is $\$12\ 800$

ii. $\text{Cost} = PS \times 2600$

Using Pythagoras:

$$\begin{aligned}PS^2 &= PR^2 + SR^2\\ &= 5^2 + 3^2\\ &= 34\\ PS &= \sqrt{34}\end{aligned}$$

$$\begin{aligned}\therefore\ \text{Cost} &= \sqrt{34} \times 2600\\ &= 15\ 160.474\ldots\\ &= \$15\ 160\ \ \text{(nearest dollar)}\end{aligned}$$

iii. Show $C = 1000\left(5 - x + 2.6\sqrt{x^2+9}\right)$

$\text{Cost} = (PQ \times 1000) + (QS \times 2600)$

$$\begin{aligned}PQ &= 5 - x\\ QS^2 &= QR^2 + SR^2\\ &= x^2 + 3^2\\ QS &= \sqrt{x^2+9}\end{aligned}$$

$$\begin{aligned}\therefore C &= (5-x)1000 + \sqrt{x^2+9}\ (2600)\\ &= 1000\left(5 - x + 2.6\sqrt{x^2+9}\right)\quad \ldots \text{ as required}\end{aligned}$$

iv. Find the MIN cost of laying the cable

$$\begin{aligned}C &= 1000\left(5 - x + 2.6\sqrt{x^2+9}\right)\\ \frac{dC}{dx} &= 1000\left(-1 + 2.6 \times \frac{1}{2} \times 2x\left(x^2+9\right)^{-\frac{1}{2}}\right)\\ &= 1000\left(-1 + \frac{2.6x}{\sqrt{x^2+9}}\right)\end{aligned}$$

MAX/MIN when $\dfrac{dC}{dx} = 0$

$$\begin{aligned}1000\left(-1 + \frac{2.6x}{\sqrt{x^2+9}}\right) &= 0\\ \frac{2.6x}{\sqrt{x^2+9}} &= 1\\ 2.6x &= \sqrt{x^2+9}\\ (2.6)^2x^2 &= x^2 + 9\\ x^2\left(2.6^2 - 1\right) &= 9\\ x^2 &= \frac{9}{5.76}\\ &= 1.5625\\ x &= 1.25\quad (x > 0)\end{aligned}$$

If $x = 1,\ \dfrac{dC}{dx} < 0$

If $x = 2,\ \dfrac{dC}{dx} > 0$

$\therefore$ MIN when $x = 1.25$

$$\begin{aligned}C &= 1000\left(5 - 1.25 + 2.6\sqrt{1.25^2+9}\right)\\ &= 1000(122)\\ &= 12\ 200\end{aligned}$$

$\therefore$ MIN cost is $\$12\ 200$ when $x = 1.25$

IMPORTANT: Tougher derivative questions often require students to deal with multiple algebraic constants. See Worked Solutions in part (iv).

MARKER'S COMMENT: Check the nature of the critical points in these type of questions. If using the first derivative test, make sure some actual values are substituted in.

v. Underwater cable now costs $1100 per km

$$\begin{aligned}\Rightarrow\ C &= 1000(5-x) + 1100\sqrt{x^2+9}\\ &= 1000\left(5 - x + 1.1\sqrt{x^2+9}\right)\\ \frac{dC}{dx} &= 1000\left(-1 + 1.1 \times \frac{1}{2} \times 2x\left(x^2+9\right)^{-\frac{1}{2}}\right)\\ &= 1000\left(-1 + \frac{1.1x}{\sqrt{x^2+9}}\right)\end{aligned}$$

MAX/MIN when $\dfrac{dC}{dx} = 0$

$$\begin{aligned}1000\left(-1 + \frac{1.1x}{\sqrt{x^2+9}}\right) &= 0\\ \frac{1.1x}{\sqrt{x^2+9}} &= 1\\ 1.1x &= \sqrt{x^2+9}\\ 1.1^2x^2 &= x^2 + 9\\ x^2\left(1.1^2 - 1\right) &= 9\end{aligned}$$

$$x^2 = \frac{9}{0.21}$$

$$x \approx 6.5 \text{ km (to 1 d.p.)}$$

$$\Rightarrow \text{ no solution since } x \leq 5$$

If we lay cable PR then RS

$$\Rightarrow \text{ Cost} = 5 \times 1100 + 3 \times 1000 = 8500$$

If we lay cable directly underwater via PS

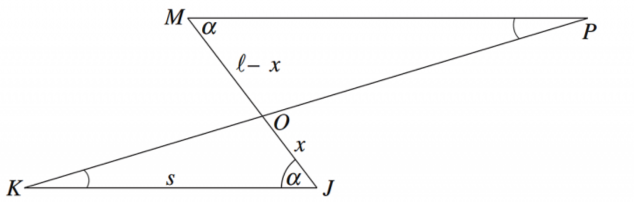
$$\Rightarrow \text{ Cost} = \sqrt{34} \times 1100 = 6414.047\ldots$$

$\therefore$ MIN cost is \$6414 by cabling directly from P to S.

MARKER'S COMMENT: Many students failed to interpret a correct calculation of $x > 5$ as providing no solution.

2. Calculus, 2ADV C3 2008 HSC 10b

i.



$A =$ Area ΔOJK + Area ΔOMP

Using sine rule

$$\text{Area } \Delta OJK = \frac{1}{2}\, xs\sin\alpha$$

Area $\Delta OMP \Rightarrow$ Need to find MP

$\angle OKJ = \angle MPO$ (alternate angles, $MP \parallel KJ$)

$\angle PMO = \angle OJK = \alpha$ (alternate angles, $MP \parallel KJ$)

$\therefore\ \Delta OJK\ |||\ \Delta OMP$ (equiangular)

$$\Rightarrow \frac{x}{s} = \frac{l-x}{MP} \quad \text{(corresponding sides of similar triangles)}$$

$$MP = \frac{l-x}{x} \cdot s$$

$$\text{Area } \Delta\ OMP = \frac{1}{2}(l-x) \cdot MP \cdot \sin\alpha$$

$$= \frac{1}{2}(l-x) \cdot \frac{(l-x)}{x} \cdot s\sin\alpha$$

$$\therefore A = \frac{1}{2}x \cdot s\sin\alpha + \frac{1}{2}(l-x) \cdot \frac{(l-x)}{x} \cdot s\sin\alpha$$

$$= s\sin\alpha\left(\frac{1}{2}x + \frac{1}{2}(l-x) \cdot \frac{(l-x)}{x}\right)$$

$$= s\sin\alpha\left(\frac{1}{2}x + \frac{(l-x)^2}{2x}\right)$$

$$= s\sin\alpha\left(\frac{1}{2}x + \frac{l^2}{2x} - l + \frac{1}{2}x\right)$$

$$= s\left(x - l + \frac{l^2}{2x}\right)\sin\alpha \quad \ldots \text{as required}$$

♦♦♦ *Low mean marks highlighted (although exact data not available before 2009).*

ii. Find x such that A is a minimum

$$A = s\left(x - l + \frac{l^2}{2x}\right)\sin\alpha$$

$$\frac{dA}{dx} = s\left(1 - \frac{l^2}{2x^2}\right)\sin\alpha$$

MARKER'S COMMENT: Students who could not complete part (i) are reminded that they can still proceed to part (ii) and attempt to differentiate the result given. Note that l and α are constants when differentiating.

MAX/MIN when $\dfrac{dA}{dx} = 0$

$$s\left(1 - \frac{l^2}{2x^2}\right)\sin\alpha = 0$$

$$\begin{aligned} \frac{l^2}{2x^2} &= 1 \\ 2x^2 &= l^2 \\ x^2 &= \frac{l^2}{2} \\ x &= \frac{l}{\sqrt{2}}, \quad x > 0 \end{aligned}$$

$$\frac{d^2A}{dx^2} = s\left(\frac{l^2}{2x^3}\right)\sin\alpha$$

Since $0 < \alpha < 90^\circ \Rightarrow \sin\alpha > 0, \ \ l > 0$ and $x > 0$

$$\frac{d^2A}{dx^2} > 0 \quad \Rightarrow \text{MIN at } x = \frac{l}{\sqrt{2}}$$

iii. Since $MP = \dfrac{(l - x)}{x}s$ and MIN when $x = \dfrac{l}{\sqrt{2}}$

$$\begin{aligned} MP &= \left(\frac{l - \frac{l}{\sqrt{2}}}{\frac{l}{\sqrt{2}}}\right)s \times \frac{\sqrt{2}}{\sqrt{2}} \\ &= \frac{\left(\sqrt{2}l - l\right)}{l}s \\ &= \left(\sqrt{2} - 1\right)s \text{ metres} \end{aligned}$$

$\therefore MP = \left(\sqrt{2} - 1\right)s$ metres when A is a MIN.

3. Calculus, 2ADV C3 2011 HSC 10b

i. Need to show $P = r(\theta + 2)$

$$P = (2 \times r) + \frac{\theta}{2\pi} \times 2\pi r$$
$$= 2r + r\theta$$
$$= r(\theta + 2) \quad \ldots \text{ as required}$$

ii. Need to show $A = \frac{1}{2}Pr - r^2$.

Since $P = r(\theta + 2) \quad \Rightarrow \theta = \dfrac{P - 2r}{r}$

$$\therefore A = \frac{1}{2}r^2\theta$$
$$= \frac{1}{2}r^2 \cdot \frac{P - 2r}{r}$$
$$= \frac{1}{2}\left(Pr - 2r^2\right)$$
$$= \frac{1}{2}Pr - r^2 \quad \ldots \text{ as required}$$

♦ Mean mark 41%.
TIP: Area of a sector $= \pi r^2 \times$ the percentage of the circle that the sector angle accounts for, i.e. $\frac{\theta}{2\pi}$ radians.
$\therefore A = \pi r^2 \times \frac{\theta}{2\pi} = \frac{1}{2}r^2\theta$.

iii. $A = \frac{1}{2}Pr - r^2$

$$\frac{dA}{dr} = \frac{1}{2}P - 2r$$

MAX or MIN when $\dfrac{dA}{dr} = 0$

$$\frac{1}{2}P - 2r = 0$$
$$2r = \frac{1}{2}P$$
$$r = \frac{P}{4}$$

$$\frac{d^2A}{dr^2} = -2 \quad \Rightarrow \text{MAX}$$

$\therefore$ Area is a MAX when $r = \dfrac{P}{4}$ units

♦♦ *Mean mark 29%*
MARKER'S COMMENT: The second derivative test proved much more successful and easily proven in this part. Make sure you are comfortable choosing between it and the 1st derivative test depending on the required calcs.

iv. Need to find θ when area is a MAX $\Rightarrow r = \dfrac{P}{4}$

$$P = r(\theta + 2)$$
$$= \frac{P}{4}(\theta + 2)$$
$$4P = P(\theta + 2)$$
$$\theta + 2 = 4$$
$$\theta = 2 \text{ radians}$$

♦♦♦ *Mean mark 20%*

v. For a sector to exist $0 < \theta < 2\pi$, and $\theta = \dfrac{P - 2r}{r}$

$$\Rightarrow \frac{P - 2r}{r} > 0$$
$$P - 2r > 0$$
$$r < \frac{P}{2}$$
$$\Rightarrow \frac{P - 2r}{r} < 2\pi$$
$$P - 2r < 2r\pi$$
$$2r\pi + 2r > P$$
$$2r(\pi + 1) > P$$
$$r > \frac{P}{2(\pi + 1)}$$

$$\therefore \frac{P}{2(\pi + 1)} < r < \frac{P}{2} \quad \ldots \text{ as required}$$

♦♦♦ Mean mark 4%. *A BEAST!*
MARKER'S COMMENT: When asked to 'explain', students should support their answer with a mathematical argument.

4. Calculus, 2ADV C3 2004 HSC 10b

i. Area $\Delta ABC = \dfrac{1}{2}\ bc\sin A$

$$\text{Area } \Delta AST = \frac{1}{2}\ xy\sin A$$

$$\text{Area } \Delta AST = \frac{1}{2} \times \text{Area } \Delta ABC \quad \text{(given)}$$

$$\frac{1}{2}\ xy\sin A = \frac{1}{2} \times \frac{1}{2}\ bc\sin A$$

$$xy\sin A = \frac{1}{2}\ bc\sin A$$

$$\therefore xy = \frac{1}{2}\ bc \quad \ldots \text{as required.}$$

ii. Using the cosine rule in ΔAST,

$$z^2 = x^2 + y^2 - 2xy\cos A$$

$$\text{Since } \ xy = \frac{1}{2}\ bc$$

$$\therefore y = \frac{bc}{2x}$$

$$\therefore z^2 = x^2 + \left(\frac{bc}{2x}\right)^2 - 2x\left(\frac{bc}{2x}\right)\cos A$$

$$= x^2 + \frac{b^2c^2}{4x^2} - bc\cos A$$

iii.

$$z^2 = x^2 + \left(\frac{b^2c^2}{4}\right)x^{-2} - bc\cos A$$

$$\frac{d(z^2)}{dx} = 2x - \frac{2b^2c^2}{4}x^{-3} = 2x - \frac{b^2c^2}{2x^3}$$

$$\frac{d^2(z^2)}{dx^2} = 2 + \frac{3b^2c^2}{2}x^{-4} = 2 + \frac{3b^2c^2}{2x^4}$$

$$\text{SP's occur when } \ \frac{d(z^2)}{dx} = 0$$

$$2x - \frac{b^2c^2}{2x^3} = 0$$

$$4x^4 - b^2c^2 = 0$$

$$4x^4 = b^2c^2$$

$$x^4 = \frac{b^2c^2}{4}$$

$$x^2 = \frac{bc}{2}$$

$$x = \sqrt{\frac{bc}{2}}, \quad (x > 0)$$

$$\frac{d^2(z^2)}{dx^2} = 2 + \frac{3b^2c^2}{2x^4} > 0 \quad \text{(for all } x\text{)}$$

$$\therefore \ \text{A minimum value of } z^2 \text{ when } x = \sqrt{\frac{bc}{2}}.$$

iv. $\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$

$$\text{When } \ x = \sqrt{\frac{bc}{2}},$$

$$\therefore z^2 = x^2 + \frac{b^2c^2}{4x^2} - bc\cos A$$

$$= \frac{bc}{2} + \frac{b^2c^2}{4\left(\frac{bc}{2}\right)} - bc\left(\frac{b^2 + c^2 - a^2}{2bc}\right)$$

$$= \frac{bc}{2} + \frac{bc}{2} - \left(\frac{b^2 + c^2 - a^2}{2}\right)$$

$$= \frac{2bc - b^2 - c^2 + a^2}{2}$$

$$= \frac{a^2 - (b^2 - 2bc + c^2)}{2}$$

$$= \frac{1}{2}\left[a^2 - (b - c)^2\right]$$

$$= \frac{1}{2}[(a - (b - c))(a + (b - c))]$$

$$= \frac{1}{2}[(a - b + c)(a + b - c)]$$

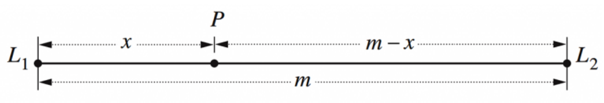
$$= \frac{1}{2}[(a + b + c - 2b)(a + b + c - 2c)]$$

$$= \frac{(P - 2b)(P - 2c)}{2} \quad (\text{using } \ P = a + b + c)$$

$$\therefore z = \sqrt{\frac{(P - 2b)(P - 2c)}{2}} \quad \text{metres} \quad \ldots \text{as required.}$$

5. Calculus, 2ADV C3 2007 HSC 10b

i.



$$N = \frac{L}{d^2}$$

$$\text{Noise from } L_1 = \frac{L_1}{x^2}$$

$$\text{Noise from } L_2 = \frac{L_2}{(m-x)^2}$$

$$\therefore N = \frac{L_1}{x^2} + \frac{L_2}{(m-x)^2}$$

ii. $N = L_1\, x^{-2} + L_2(m{-}x)^{-2}$

$$\frac{dN}{dx} = -\,2L_1x^{-3} + -\,2L_2(m{-}x)^{-3} \times \frac{d}{dx}(m{-}x)$$

$$= \frac{-\,2L_1}{x^3} + \frac{2L_2}{(m{-}x)^3}$$

$$\text{Max or min when } \frac{dN}{dx} = 0$$

$$\frac{2L_1}{x^3} = \frac{2L_2}{(m{-}x)^3}$$

$$2L_1(m{-}x)^3 = 2L_2\, x^3$$

$$L_1(m{-}x)^3 = L_2\, x^3$$

$$\sqrt[3]{L_1}(m{-}x) = \sqrt[3]{L_2}\, x$$

$$\sqrt[3]{L_1}\, m{-}\sqrt[3]{L_1}\, x = \sqrt[3]{L_2}\, x$$

$$\sqrt[3]{L_2}\, x + \sqrt[3]{L_1}\, x = \sqrt[3]{L_1}\, m$$

$$x\left(\sqrt[3]{L_2} + \sqrt[3]{L_1}\right) = \sqrt[3]{L_1}\, m$$

$$x = \frac{\sqrt[3]{L_1}\, m}{\left(\sqrt[3]{L_2} + \sqrt[3]{L_1}\right)}$$

$$\frac{dN}{dx} = -\,2L_1\, x^{-3} + 2L_2(m{-}x)^{-3}$$

$$\frac{d^2N}{dx^2} = 6L_1\, x^{-4} {-} 6L_2(m{-}x)^{-4} \times -\,1$$

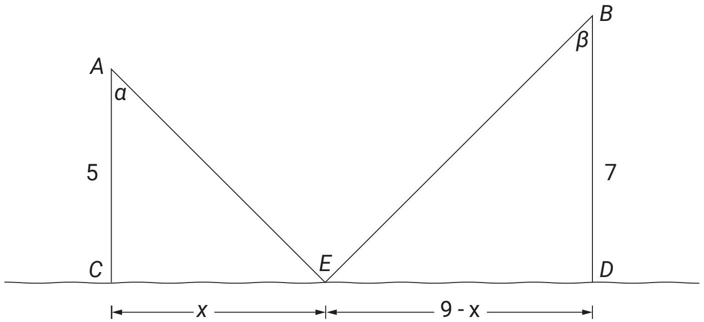
$$= \frac{6L_1}{x^4} + \frac{6L_2}{(m{-}x)^4} > 0$$

$\therefore$ A minimum occurs when

$$x = \frac{\sqrt[3]{L_1}\, m}{\left(\sqrt[3]{L_2} + \sqrt[3]{L_1}\right)}$$

6. Calculus, 2ADV C3 2017 HSC 16a

i.



Using Pythagoras:

$L = AE + EB$

$= \sqrt{5^2 + x^2} + \sqrt{7^2 + (9 - x)^2}$

$= \sqrt{25 + x^2} + \sqrt{49 + (9 - x)^2}$ ... as required

ii. From diagram:

$\sin\alpha = \dfrac{x}{\sqrt{25 + x^2}}$ and $\sin\beta = \dfrac{9 - x}{\sqrt{49 + (9 - x)^2}}$

♦ *Mean mark 41%.*

$L = \sqrt{25 + x^2} + \sqrt{49 + (9 - x)^2}$

$\dfrac{dL}{dx} = \dfrac{2x}{\sqrt{25 + x^2}} - \dfrac{2(9 - x)}{\sqrt{49 + (9 - x)^2}}$

If $\dfrac{dL}{dx} = 0$,

$\Rightarrow \dfrac{2x}{\sqrt{25 + x^2}} = \dfrac{2(9 - x)}{\sqrt{49 + (9 - x)^2}}$

$\dfrac{x}{\sqrt{25 + x^2}} = \dfrac{9 - x}{\sqrt{49 + (9 - x)^2}}$

$\sin\alpha = \sin\beta$ ... as required

iii. If $\sin\alpha = \sin\beta$, then $\alpha = \beta$ and

$\Delta ACE \mathrel{|||} \Delta BDE$

♦♦ *Mean mark 29%.*

Using corresponding sides of similar triangles:

$\dfrac{x}{5} = \dfrac{9 - x}{7}$

$7x = 45 - 5x$

$12x = 45$

$\therefore x = \dfrac{45}{12}$ km

♦♦♦ *Mean mark 9%.*

iv. If point B is reflected across the river, AEB will be a straight line.
If any other point is chosen, AEB would not be straight and the distance would be longer.

7. Calculus, 2ADV C3 SM-Bank 13

i. In ΔBCD,

$$\cos\theta = \frac{BD}{a}$$
$$\therefore BD = a\cos\theta$$
$$\sin\theta = \frac{CD}{a}$$
$$\therefore CD = a\sin\theta$$

ii. $L = 4 + 2BD + CD + a$

$$= 4 + 2a\cos\theta + a\sin\theta + a$$
$$= 4 + a + 2a\cos\theta + a\sin\theta$$

iii. Noting that a is a constant:

♦ *Mean mark (Vic) 35%.*

$$\frac{dL}{d\theta} = -2a\sin\theta + a\cos\theta$$

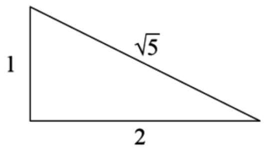
When $\dfrac{dL}{d\theta} = 0$,

$$-2a\sin\theta + a\cos\theta = 0$$
$$a\cos\theta = 2a\sin\theta$$
$$\therefore\ BD = 2CD \quad \text{(using part (a))}$$

iv. SP's when $\dfrac{dL}{d\theta} = 0$,

♦♦♦ *Mean mark (Vic) 5%.*

$$-2a\sin\theta + a\cos\theta = 0$$
$$\sin\theta = \frac{1}{2}\cos\theta$$
$$\tan\theta = \frac{1}{2}$$



If $\tan\theta = \dfrac{1}{2}$, $\cos\theta = \dfrac{2}{\sqrt{5}}$, $\sin\theta = \dfrac{1}{\sqrt{5}}$

$$L_{\max} = 4 + a + 2a\cos\theta + a\sin\theta$$
$$= 4 + \left(3\sqrt{5}\right) + 2\left(3\sqrt{5}\right)\left(\frac{2}{\sqrt{5}}\right) + \left(3\sqrt{5}\right)\left(\frac{1}{\sqrt{5}}\right)$$
$$= 4 + 3\sqrt{5} + 12 + 3$$
$$= 19 + 3\sqrt{5}\text{ cm}$$

8. Calculus, 2ADV C3 2024 HSC 31

a. Area $=$ Sector OQR $-$ Sector OTS

COMMENT: Sector/arc calculations used in solution are for those who don't want to remember formulas.

$$A = \frac{\theta}{2\pi} \times \pi(1+x)^2 - \frac{\theta}{2\pi} \times \pi \times 1^2$$

$$A = \frac{\theta}{2}(1+x)^2 - \frac{\theta}{2}$$

$$A = \frac{\theta}{2}(1 + 2x + x^2 - 1)$$

$$A = \frac{\theta}{2}x(x+2)$$

$$\theta = \frac{2A}{x(x+2)}$$

$$\begin{aligned}\text{Perimeter } QRST &= \frac{\theta}{2\pi} \times 2\pi(x+1) + \frac{\theta}{2\pi} \times 2\pi(1) + 2x \\ &= \theta(x+1) + \theta + 2x \\ &= \theta(x+2) + 2x \\ &= \frac{2A}{x(x+2)}(x+2) + 2x \\ &= 2x + \frac{2A}{x}\end{aligned}$$

b. $P(x) = 2x + \dfrac{2A}{x}$

$$P'(x) = 2 - \frac{2A}{x^2}$$

$$P''(x) = \frac{4A}{x^3} > 0 \ \ (x, A > 0)$$

$\Rightarrow$ MIN when $P'(x) = 0$:

$$2 - \frac{2A}{x^2} = 0$$

$$2x^2 = 2A$$

$$x^2 = A$$

Using part (a):

$$\begin{aligned}\theta &= \frac{2A}{x(x-2)} \\ &= \frac{2x^2}{x(x+2)} \\ &= \frac{2x}{x+2} \\ &= 2 \times \frac{x}{x+2} \\ &< 2 \ \ \left(\text{since } \frac{x}{x+2} < 1 \ \text{ for all } x\right)\end{aligned}$$

$\therefore$ If $P(x)$ is minimised, $\theta < 2$.

9. Calculus, 2ADV C4 2019 HSC 16c

i. $y = x^r$

$$\frac{dy}{dx} = rx^{r-1}$$

When $x = 1,\ \frac{dy}{dx} = r$

Equation of tangent:

$$y-1 = r(x-1)$$
$$y = rx-r+1$$

When $y = 0$:

$$rx-r+1 = 0$$
$$rx = r-1$$
$$x = \frac{r-1}{r}$$

$\therefore$ Tangent meets x-axis at $\left(\frac{r-1}{r}, 0\right)$

♦♦ *Mean mark part (i) 31%.*

ii. Area under curve

$$= \int_0^1 x^r$$
$$= \left[\frac{1}{r+1} \cdot x^{r+1}\right]_0^1$$
$$= \frac{1}{r+1} \times 1^{r+1}-0$$
$$= \frac{1}{r+1}$$

Area under tangent

$$= \frac{1}{2} \times b \times h$$
$$= \frac{1}{2}\left(1-\frac{r-1}{r}\right) \times 1$$
$$= \frac{1}{2}\left(1-\frac{r-1}{r}\right)$$

$\therefore$ R`= 1/(r + 1) − 1/2(1 − (r − 1)/r)"

$$= \frac{1}{r+1}-\frac{1}{2r}[r-(r-1)]$$
$$= \frac{1}{r+1}-\frac{1}{2r}$$
$$= \frac{2r-(r+1)}{2r(r+1)}$$
$$= \frac{r-1}{2r(r+1)}$$

♦♦♦ *Mean mark part (ii) 21%.*

iii. $R = \frac{r-1}{2r(r+1)} = \frac{r-1}{2r^2+2r}$

$$\frac{dR}{dr} = \frac{(2r^2+2r) \times 1-(r-1)(4r+2)}{(2r^2+2r)^2}$$
$$= \frac{2r^2+2r-4r^2-2r+4r+2}{(2r^2+2r)^2}$$
$$= \frac{-2r^2+4r+2}{(2r^2+2r)^2}$$
$$= \frac{-2(r^2-2r-1)}{(2r^2+2r)^2}$$

Find r when $\frac{dR}{dr} = 0$:

$$r^2-2r-1 = 0$$
$$r = \frac{2 \pm \sqrt{(-2)^2-4 \times 1 \times (-1)}}{2}$$
$$= \frac{2 \pm \sqrt{8}}{2}$$
$$= 1+\sqrt{2}\ \ (r \geq 1)$$

r	1	$1+\sqrt{2}$	3
$\frac{dR}{dr}$	$\frac{1}{4}$	0	$-\frac{1}{144}$

$\therefore R_{\max}$ occurs when $r = 1+\sqrt{2}$

♦♦ *Mean mark part (iii) 23%.*

10. Calculus, 2ADV C3 2022 HSC 31

a. Show $y = \dfrac{2x}{x-1}$

Since $m_{YP} = m_{PX}$:

$$\frac{y-2}{0-1} = \frac{2-0}{1-x}$$

$$y - 2 = \frac{-2}{1-x}$$

$$y = 2 - \frac{2}{1-x}$$

$$= \frac{2(1-x)-2}{1-x}$$

$$= \frac{-2x}{1-x}$$

$$= \frac{2x}{x-1} \quad \ldots \text{as required}$$

♦♦♦ Mean mark part (a) 17%.

COMMENT: $y = \dfrac{2x}{x-1}$ is the expression of a relationship between the intercepts and not the equation of the line.

b. $A = \dfrac{1}{2} \times b \times h$

$$= \frac{1}{2}x\left(\frac{2x}{x-1}\right)$$

$$= \frac{x^2}{x-1}$$

♦♦ Mean mark part (b) 29%.

$$\frac{dA}{dx} = \frac{(x-1)\cdot 2x - x^2(1)}{(x-1)^2}$$

$$= \frac{2x^2 - 2x - x^2}{(x-1)^2}$$

$$= \frac{x(x-2)}{(x-1)^2}$$

SP's occur when $\dfrac{dA}{dx} = 0$:

$x = 0$ or 2

Use 1st derivative test to find max/min:

x	-1	0	1.5	2	3
$\frac{dA}{dx}$	$\frac{3}{4}$	0	-3	0	$\frac{3}{4}$
	+	0	−	0	+

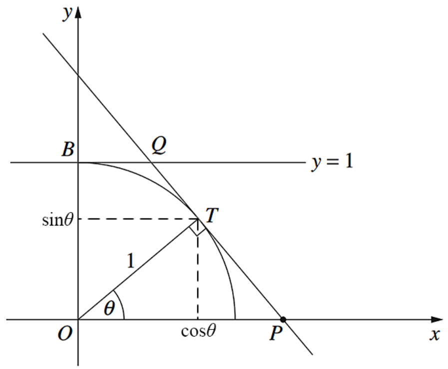
$\Rightarrow$ MIN at $x = 2$

$$\therefore A_{\text{min}} = \frac{1}{2} \times 2 \times \frac{2 \times 2}{2-1}$$

$$= 4 \text{ u}^2$$

11. Calculus, 2ADV C3 2012 HSC 16b

i.



Find T:

Since $\cos\theta = \dfrac{x}{1}$ and $\sin\theta = \dfrac{y}{1}$

$\therefore T\ (\cos\theta, \sin\theta)$

Gradient of $OT = \dfrac{\sin\theta}{\cos\theta}$

$\therefore$ Gradient $PT = -\dfrac{\cos\theta}{\sin\theta}$ ($\perp$ lines)

Equation of PT where

$m = -\dfrac{\cos\theta}{\sin\theta}$, and through $(\cos\theta, \sin\theta)$

Using $y - y_1 = m(x - x_1)$

$$y - \sin\theta = -\frac{\cos\theta}{\sin\theta}(x - \cos\theta)$$

$$y\sin\theta - \sin^2\theta = -x\cos\theta + \cos^2\theta$$

$$x\cos\theta + y\sin\theta = \sin^2\theta + \cos^2\theta$$

$$x\cos\theta + y\sin\theta = 1 \quad \ldots \text{as required}$$

♦♦ *Mean mark 20%*

ii. Find Q:

$Q \Rightarrow$ intersection of $x\cos\theta + y\sin\theta = 1$ and $y = 1$

$$x\cos\theta + \sin\theta = 1$$

$$x\cos\theta = 1 - \sin\theta$$

$$x = \frac{1 - \sin\theta}{\cos\theta}$$

$\therefore$ Length of BQ is $\dfrac{1 - \sin\theta}{\cos\theta}$ units

iii. Show Area $OPQB = \dfrac{2 - \sin\theta}{2\cos\theta}$

$A = \dfrac{1}{2}h(a+b)$ where $h = OB = 1$ $a = OP$ and

$$b = BQ = \frac{1 - \sin\theta}{\cos\theta}$$

Find length OP:

$P \Rightarrow x\cos\theta + y\sin\theta = 1$ cuts x-axis

$$x\cos\theta = 1$$

$$x = \frac{1}{\cos\theta}$$

$\Rightarrow$ Length $OP = \dfrac{1}{\cos\theta}$

♦♦ *Mean mark 24%*

$$\text{Area } OPQB = \frac{1}{2} \times 1\left(\frac{1}{\cos\theta} + \frac{1 - \sin\theta}{\cos\theta}\right)$$

$$= \frac{1}{2}\left(\frac{2 - \sin\theta}{\cos\theta}\right)$$

$$= \frac{2 - \sin\theta}{2\cos\theta} \text{ u}^2 \quad \ldots \text{as required}$$

iv. Find θ such that Area $OPQB$ is a MIN

$$A = \frac{2 - \sin\theta}{2\cos\theta}$$

$$\frac{dA}{d\theta} = \frac{2\cos\theta(-\cos\theta) - (2 - \sin\theta)(-2\sin\theta)}{4\cos^2\theta}$$

$$= \frac{4\sin\theta - 2\sin^2\theta - 2\cos^2\theta}{4\cos^2\theta}$$

$$= \frac{4\sin\theta - 2\left(\sin^2\theta + \cos^2\theta\right)}{4\cos^2\theta}$$

$$= \frac{4\sin\theta - 2}{4\cos^2\theta}$$

$$= \frac{2\sin\theta - 1}{2\cos^2\theta}$$

Mean mark 19%
IMPORTANT: Look for any opportunity to use the identity $\sin^2\theta + \cos^2\theta = 1 \rightarrow$ often a key to simplifying difficult trig equations.

MAX or MIN when $\dfrac{dA}{d\theta} = 0$

$$\Rightarrow 2\sin\theta - 1 = 0$$

$$\sin\theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6} \quad 0 < \theta < \frac{\pi}{2}$$

Test for MAX/MIN

If $\theta = \dfrac{\pi}{12}$ $\dfrac{dA}{d\theta} < 0$

IMPORTANT: Is the 1st or 2nd derivative test easier here? Examiners have often made one significantly easier than the other.

If $\theta = \dfrac{\pi}{3}$ $\dfrac{dA}{d\theta} > 0 \Rightarrow \text{MIN}$

$\therefore$ Area $OPQB$ is a MIN when $\theta = \dfrac{\pi}{6}$.

12. Calculus, 2ADV C3 2013 HSC 14b

i. Need to show $r^2 = 12\ 900t^2 - 18\ 000t + 10\ 000$

♦♦ *Mean mark 26%*

$RB = 100 - 50t$

$RA = 80t$

Using the cosine rule

$$\begin{aligned} r^2 &= (RB)^2 + (RA)^2 - 2(RB)(RA)\cos\angle R \\ &= (100-50t)^2 + (80t)^2 - 2(100-50t)(80t)\cos 60 \\ &= 10\ 000 - 10\ 000t + 2500t^2 + 6400t^2 - 8000t + 4000t^2 \\ &= 12\ 900t^2 - 18\ 000t + 10\ 000 \quad \ldots \text{as required} \end{aligned}$$

ii. Max/min when $\dfrac{dr^2}{dt} = 0$

♦♦ *Mean mark 27%*
ALGEBRA TIP: Finding the derivative of r^2 (rather than making r the subject), makes calculations ***much easier***. *ENSURE* you apply the test to confirm a minimum.

$$\frac{dr^2}{dt} = 25\ 800t - 18\ 000 = 0$$

$$t = \frac{18\ 000}{25\ 800} = \frac{30}{43}$$

When $t = \dfrac{30}{43}$, $\dfrac{d^2(r^2)}{dt^2} = 25\ 800 > 0 \Rightarrow \text{MIN}$

$\therefore$ Minimum distance when $t = \dfrac{30}{43}$ hr

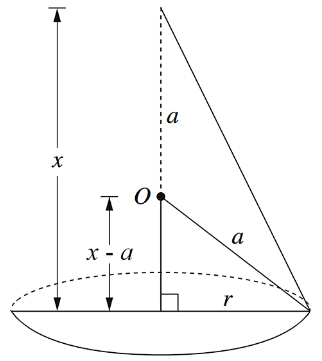
Find r when $t = \dfrac{30}{43}$

$$\begin{aligned} r^2 &= 12\ 900\left(\frac{30}{43}\right)^2 - 18\ 000\left(\frac{30}{43}\right) + 10\ 000 \\ &= 3720.9302\ldots \\ \therefore r &= \sqrt{3702.9302\ldots} \\ &\approx 60.99942\ldots \\ &\approx 61 \text{ km (nearest km)} \end{aligned}$$

$\therefore$ MIN distance between the cars is 61 km.

13. Calculus, 2ADV C3 2006 HSC 9c

i. Show $V = \dfrac{1}{3}\pi\left(2ax^2 - x^3\right)$



$V = \dfrac{1}{3}\pi r^2 h$

Using Pythagoras

$$\begin{aligned} (x-a)^2 + r^2 &= a^2 \\ r^2 &= a^2 - (x-a)^2 \\ &= a^2 - x^2 + 2ax - a^2 \\ &= 2ax - x^2 \end{aligned}$$

$$\begin{aligned} \therefore V &= \frac{1}{3} \times \pi \times \left(2ax - x^2\right) \times x \\ &= \frac{1}{3}\pi\left(2ax^2 - x^3\right) \ \ldots \text{as required} \end{aligned}$$

ii. $\dfrac{dV}{dx} = \dfrac{1}{3}\pi\left(4ax - 3x^2\right)$

$\dfrac{d^2V}{dx^2} = \dfrac{1}{3}\pi(4a - 6x)$

Max or min when $\dfrac{dV}{dx} = 0$

$$\begin{aligned} \frac{1}{3}\pi\left(4ax - 3x^2\right) &= 0 \\ 4ax - 3x^2 &= 0 \\ x(4a - 3x) &= 0 \\ 3x = 4a, &\quad x \neq 0 \\ x &= \frac{4}{3}a \end{aligned}$$

When $x = \dfrac{4}{3}a$

$$\frac{d^2V}{dx^2} = \frac{1}{3}\pi\left(4a - 6 \times \frac{4}{3}a\right)$$

$$= \frac{1}{3}\pi(-4a) < 0$$

$\Rightarrow$ MAX

$\therefore$ Cone volume is a maximum when $x = \frac{4}{3}a$.

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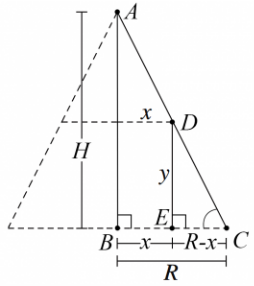
14. Calculus, 2ADV C3 2015 HSC 16c

i. Show $V = \frac{H}{R}\pi x^2(R - x)$

◆◆ Mean mark (i) 16%.



Consider ΔABC and ΔDEC

$\angle ABC = \angle DEC = 90^\circ$

$\angle BCA$ is common

$\therefore \Delta ABC \ ||| \ \Delta DEC$ (equiangular)

$\therefore \frac{DE}{EC} = \frac{AB}{BC}$ (corresponding sides of similar triangles)

$$\frac{y}{R - x} = \frac{H}{R}$$

$$y = \frac{H}{R}(R - x)$$

Volume of cylinder

$$= \pi r^2 h$$

$$= \pi x^2 \times \frac{H}{R}(R - x)$$

$$= \frac{H}{R}\pi x^2(R - x) \quad \ldots \text{ as required.}$$

ii. $V = \frac{H}{R}\pi x^2(R - x)$

◆◆ Mean mark (ii) 21%.

$$\frac{dV}{dx} = \frac{H}{R}\pi\left[\left(x^2 \times -1\right) + 2x(R - x)\right]$$

$$= \frac{H}{R}\pi\left[-x^2 + 2xR - 2x^2\right]$$

$$= \frac{H}{R}\pi\left[2xR - 3x^2\right]$$

$$\frac{dV^2}{dx^2} = \frac{H}{R}\pi[2R - 6x]$$

Max or min when $\frac{dV}{dx} = 0$

$$\frac{H}{R}\pi\left[2xR - 3x^2\right] = 0$$

$$2xR - 3x^2 = 0$$

$$x(2R - 3x) = 0$$

$$x = 0 \quad \text{or} \quad 3x = 2R$$

$$x = \frac{2R}{3}$$

When $x = 0$:

$$\frac{d^2V}{dx^2} = \frac{H}{R}\pi[2R - 0] > 0 \quad \Rightarrow \quad \text{MIN}$$

When $x = \dfrac{2R}{3}$:

$$\frac{d^2V}{dx^2} = \frac{H}{R}\pi\left[2R - 6 \times \frac{2R}{3}\right]$$

$$= \frac{H}{R}\pi[-2R] < 0 \quad \Rightarrow \text{MAX}$$

Maximum Volume of cylinder

$$= \frac{H}{R}\pi\left(\frac{2R}{3}\right)^2\left(R - \frac{2R}{3}\right)$$

$$= \frac{H}{R}\pi \times \frac{4R^2}{9} \times \frac{R}{3}$$

$$= \frac{4\pi HR^2}{27}$$

Volume of cone $= \dfrac{1}{3}\pi R^2 H$

$$\therefore \frac{\text{Max Volume of Cylinder}}{\text{Volume of cone}}$$

$$= \frac{\frac{4\pi HR^2}{27}}{\frac{1}{3}\pi R^2 H}$$

$$= \frac{4\pi HR^2}{27} \times \frac{3}{\pi R^2 H}$$

$$= \frac{4}{9}$$

$\therefore$ The volume of the inscribed cylinder does not exceed $\dfrac{4}{9}$ of the cone volume.